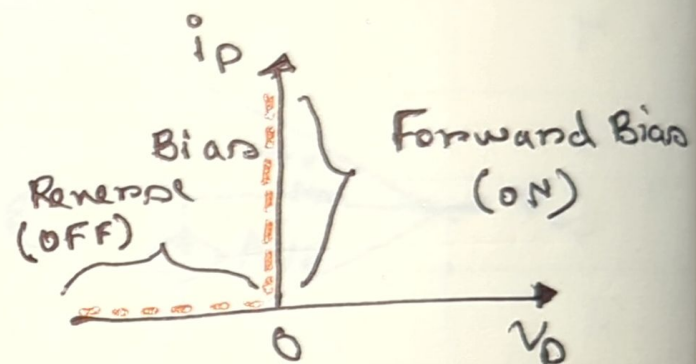
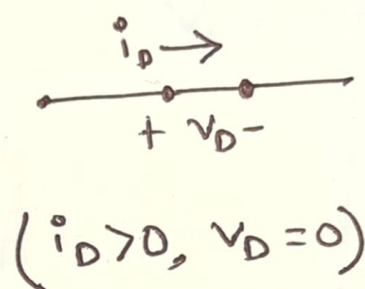
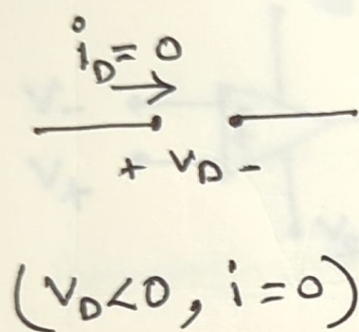


Tue Aug 5 11:42 PM

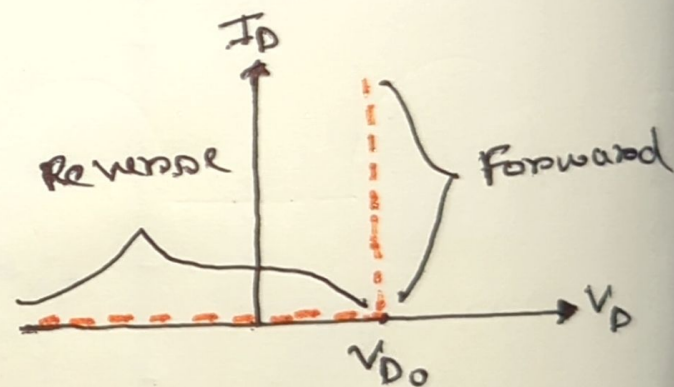
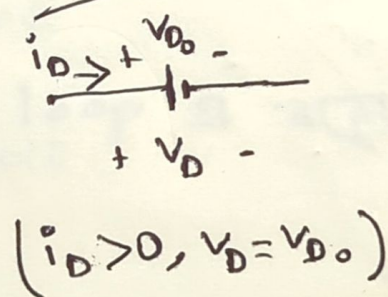
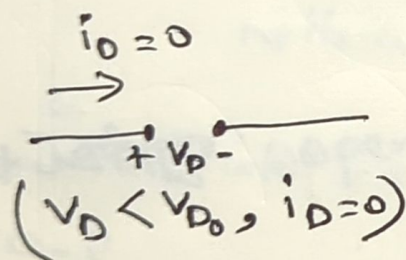
Rectifiers

Diode Models:

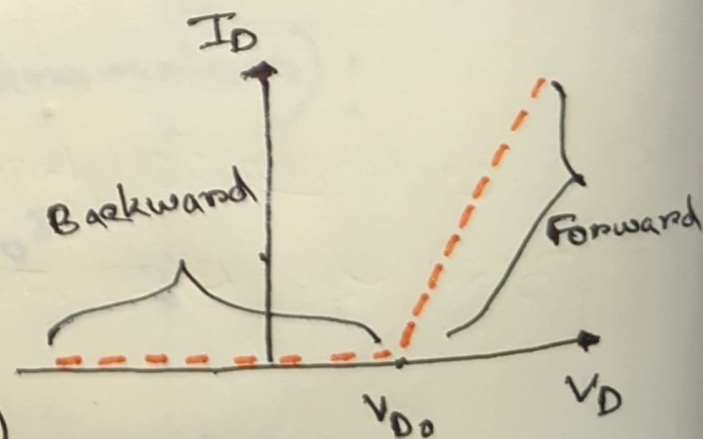
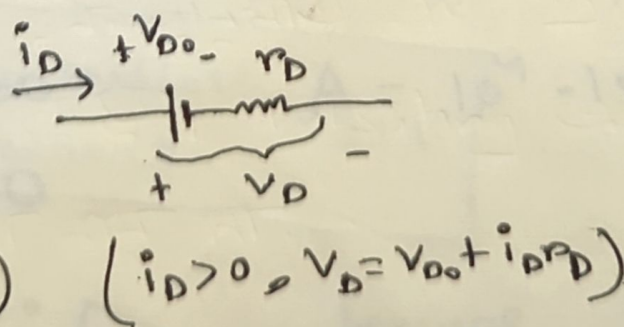
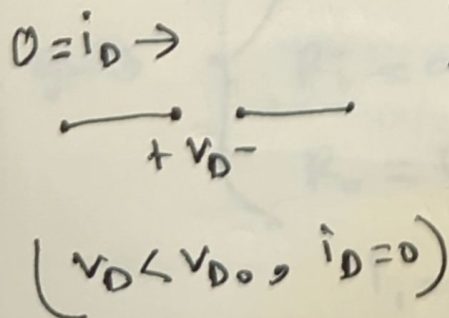
Ideal



eV D



eV D + R

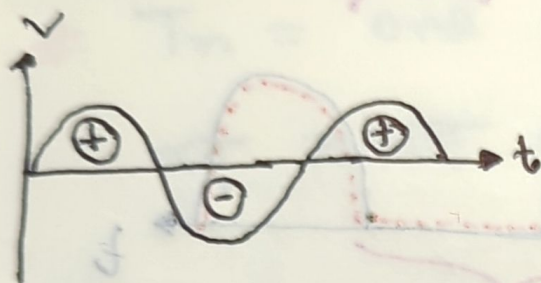


Rectifier এর কাজ কি?

= AC signal কে DC হতে পরিণত করা।

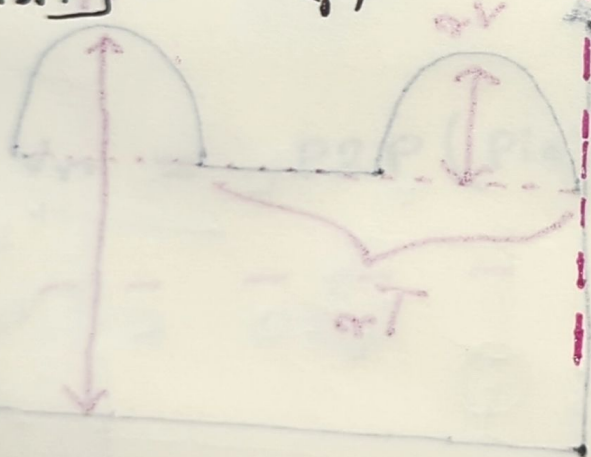
AC signal আর DC signal কি?

AC signal

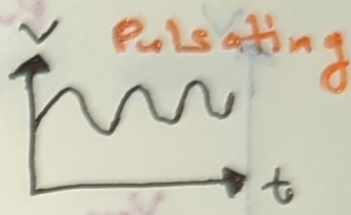
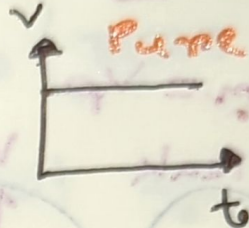
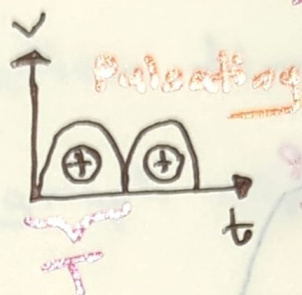


+ এবং - চিহ্ন

Polarity থাকবে,

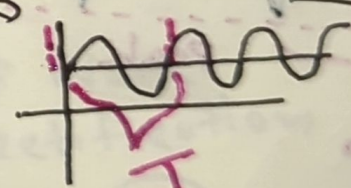


DC signal



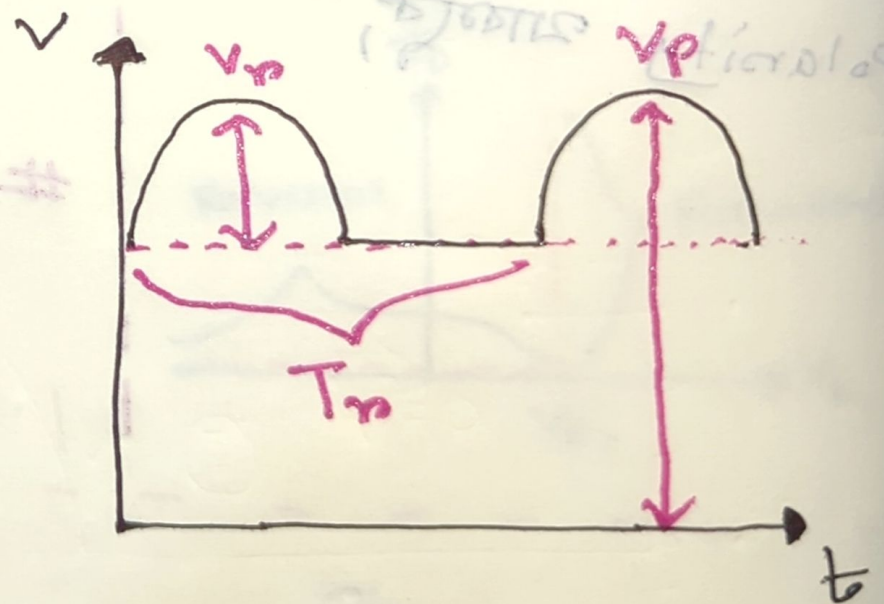
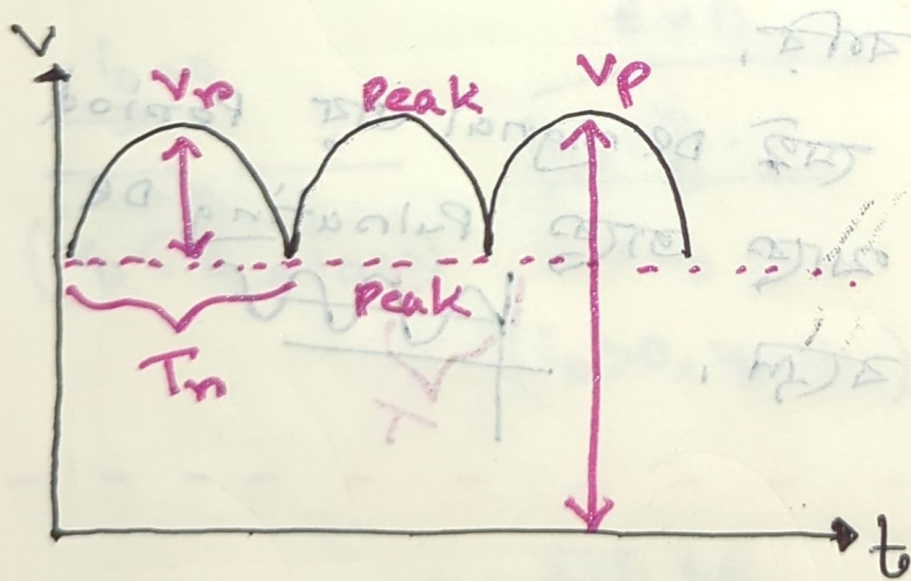
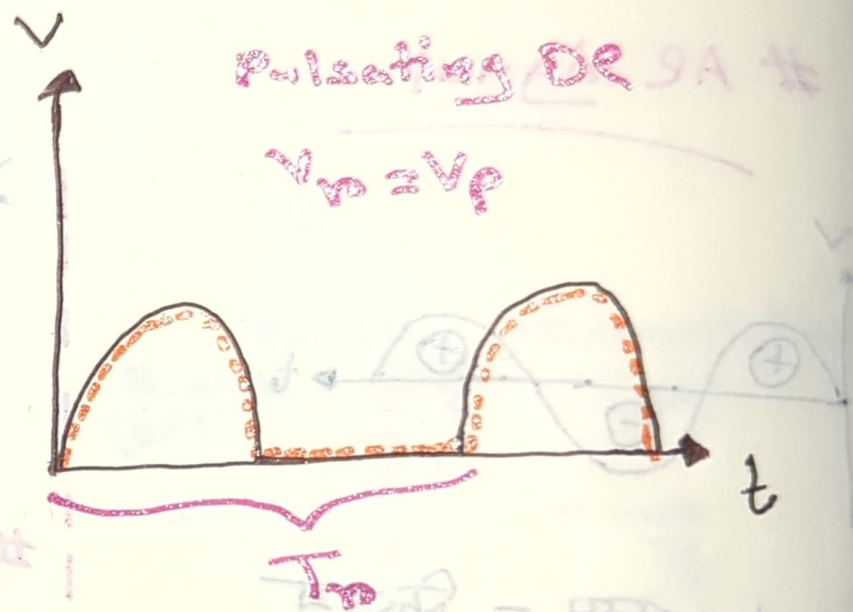
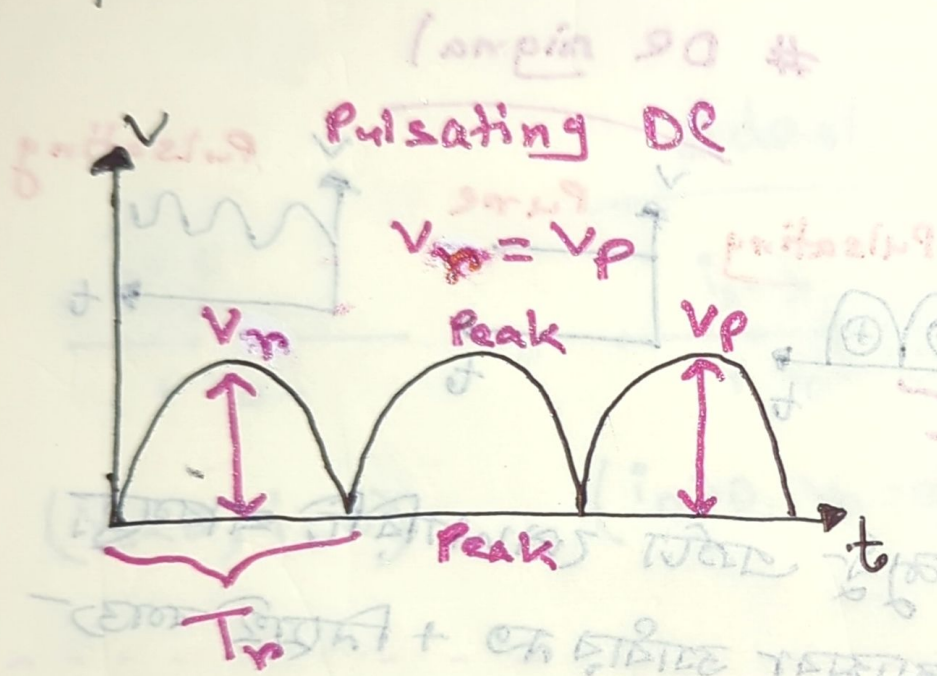
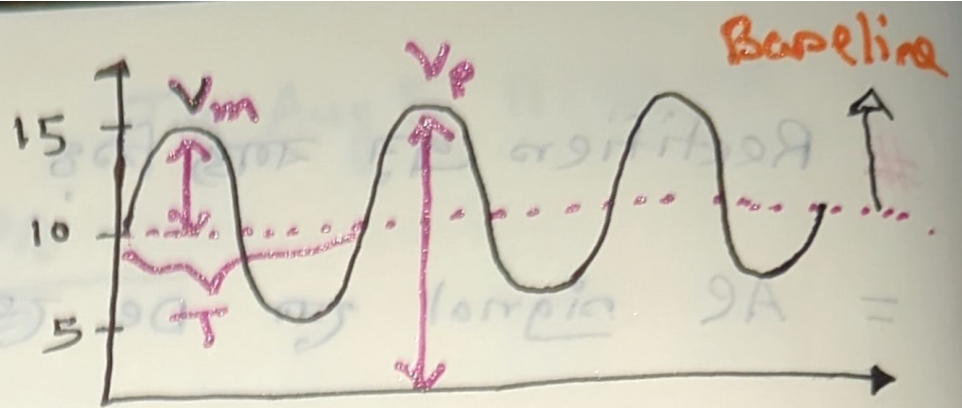
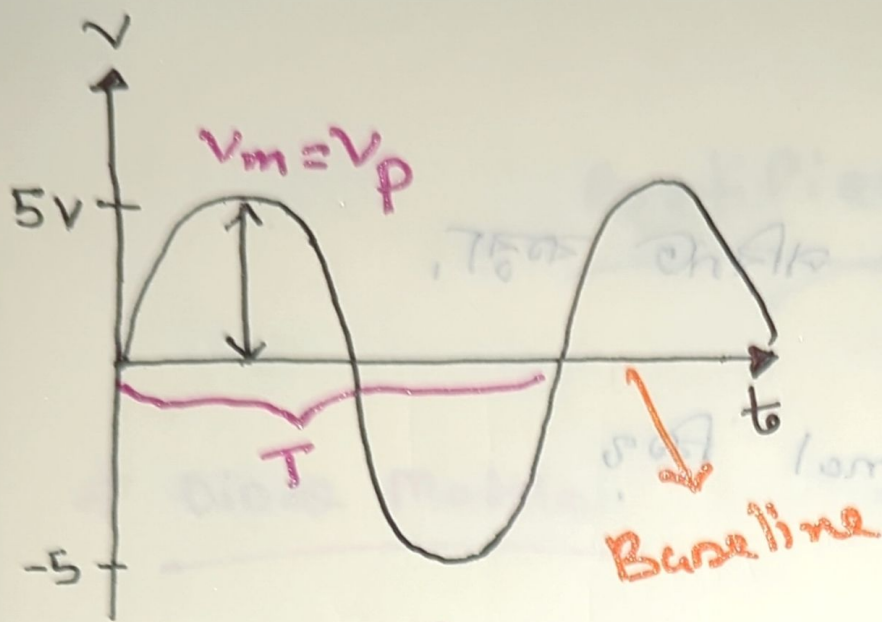
কুর্খু একটি পোলারিটি থাকবে, আমরা আরও + নিয়ে কাজ করি,

যেহে DC signal এর Period থাকে তাহলে Pulsating DC বলে।



আমরা এখানে Phase(ϕ) ও V_{DC} (Offset)

নিয়ে কাজ করব না।

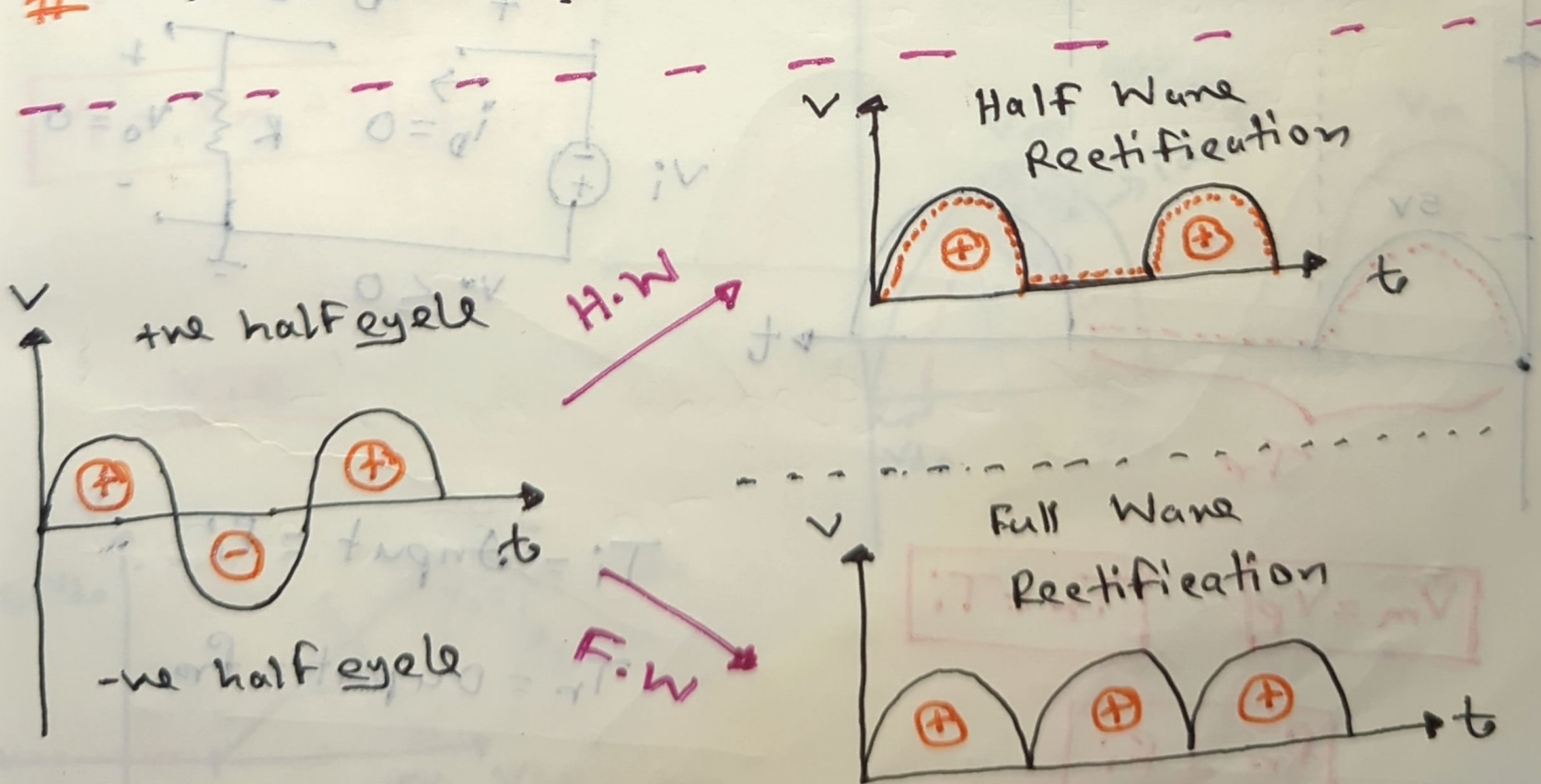


Baseline हम वृद्धांत AC signal / DC signal by offsetting AC कर सकते हैं। Pulsating कर सकते हैं और दिए गए समान रूप में है।

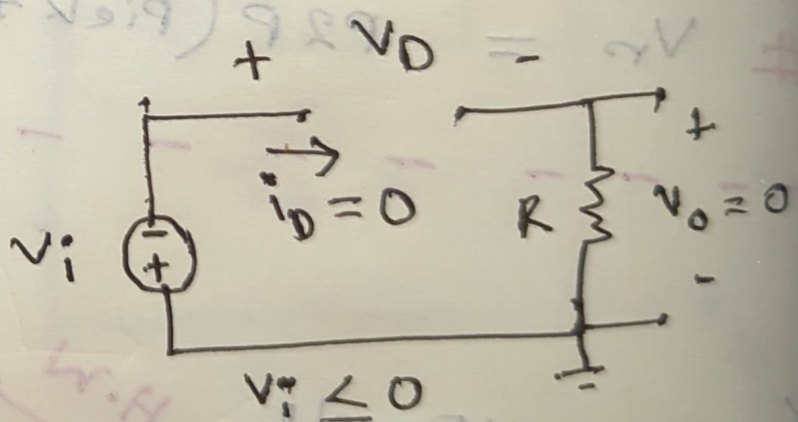
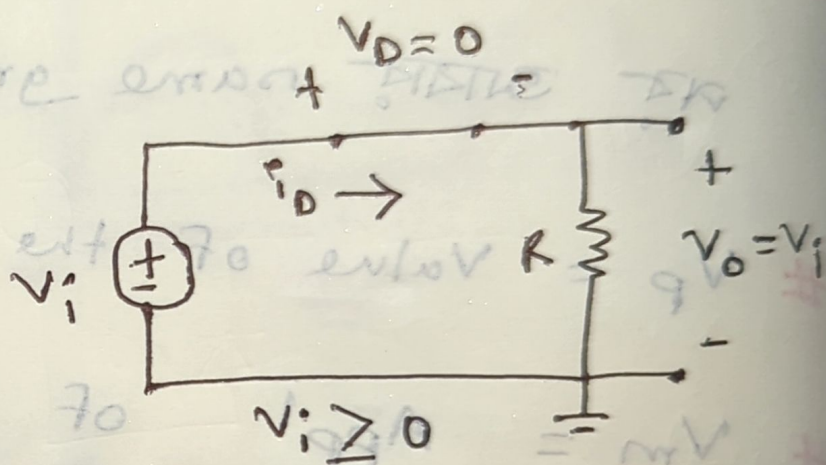
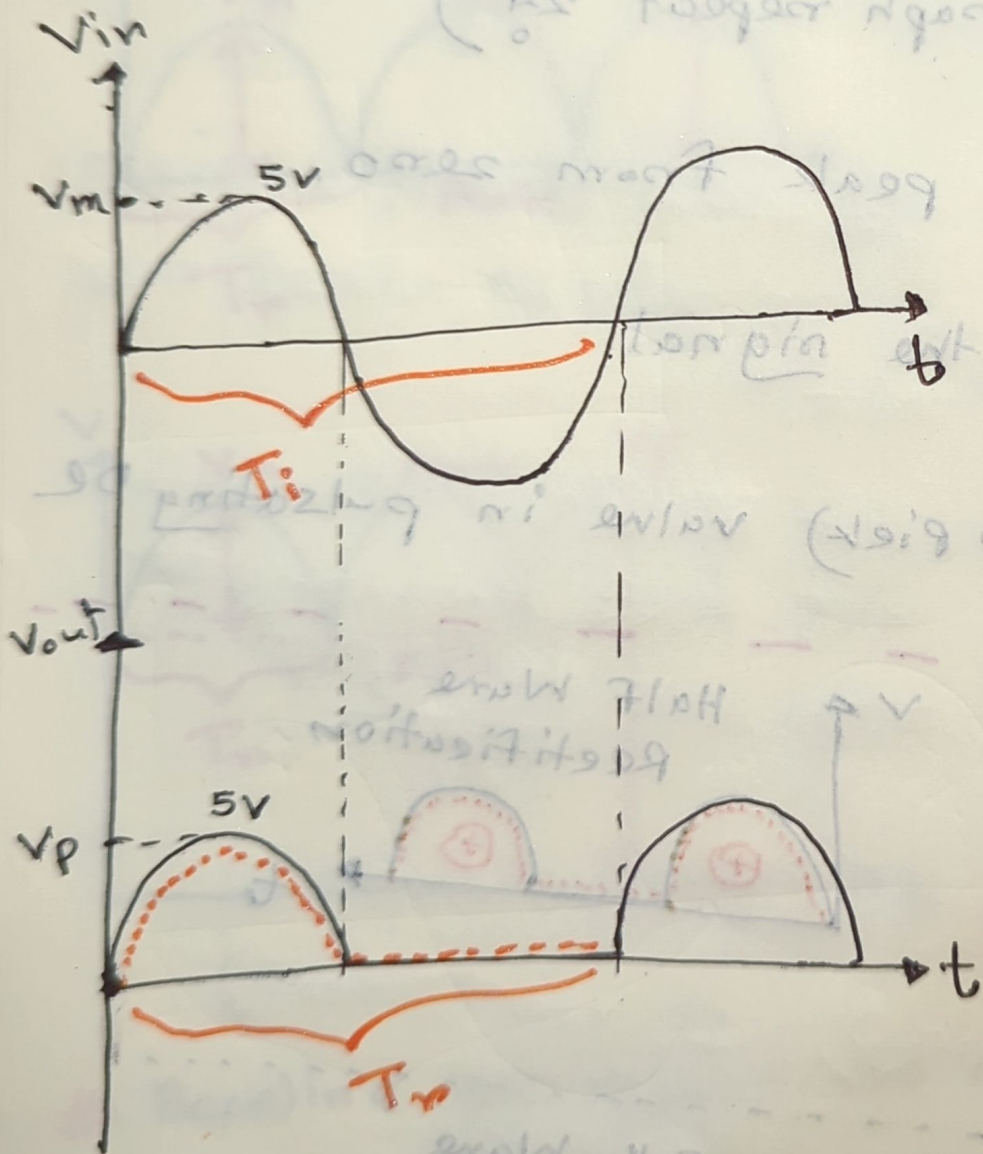
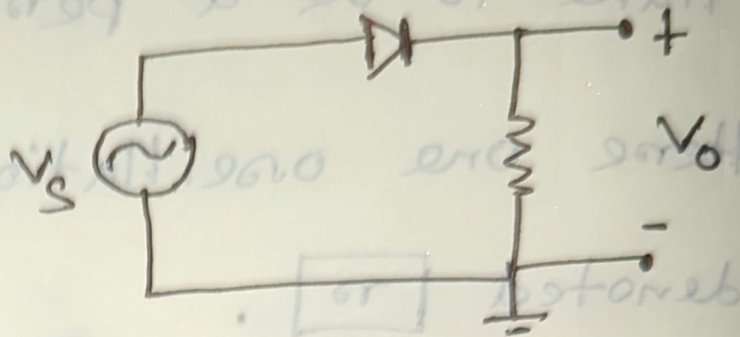
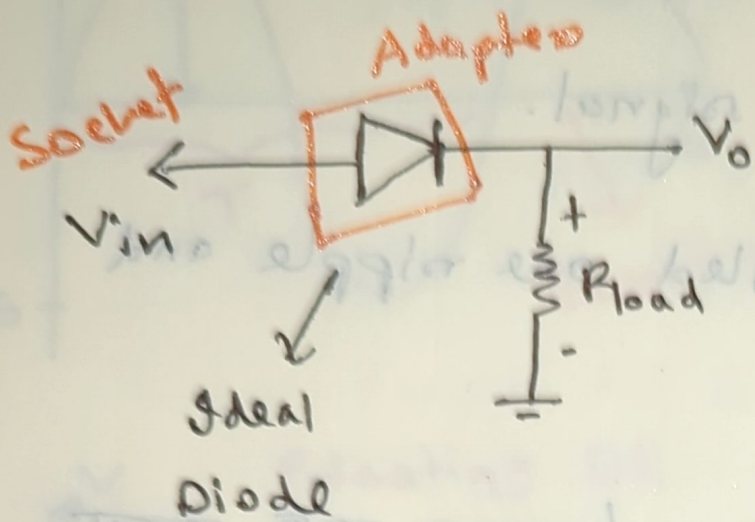
$$\text{Baseline} = \frac{\text{max} + \text{min}}{2}$$

Pulsating DC

- # Have to be a periodic signal.
- # Here one oscillation called one ripple and denoted T_r .
- # T_r = one oscillation time (মূলত কত সময় পরে আবার same graph repeat হয়)
- # V_p = Value of the peak from zero.
- # V_m = Appl of the signal
- # V_r = P2P (Peak to Peak) value in pulsating DC



H.W



$V_m = V_p$

$T_r = T_i$

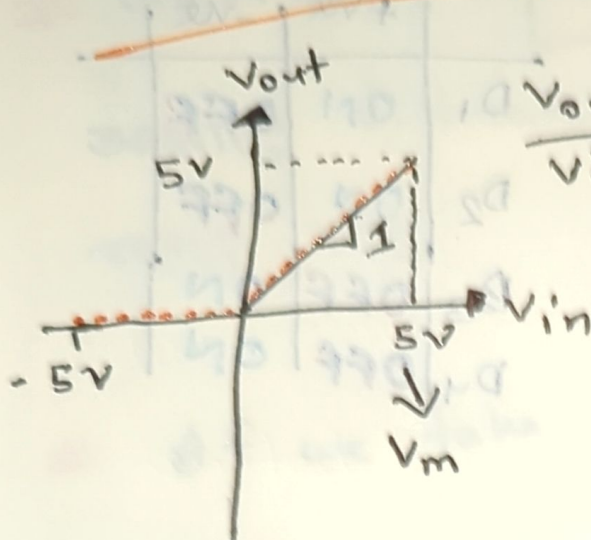
$f_r = f_i$

$T_i = \text{Input} = f_i$

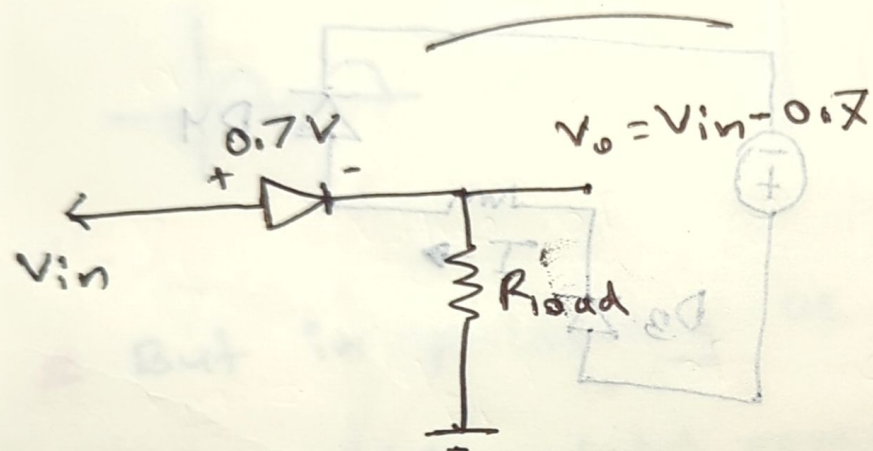
$T_r = \text{Output} = f_r$

$f \rightarrow \text{frequency}$

VTE graph



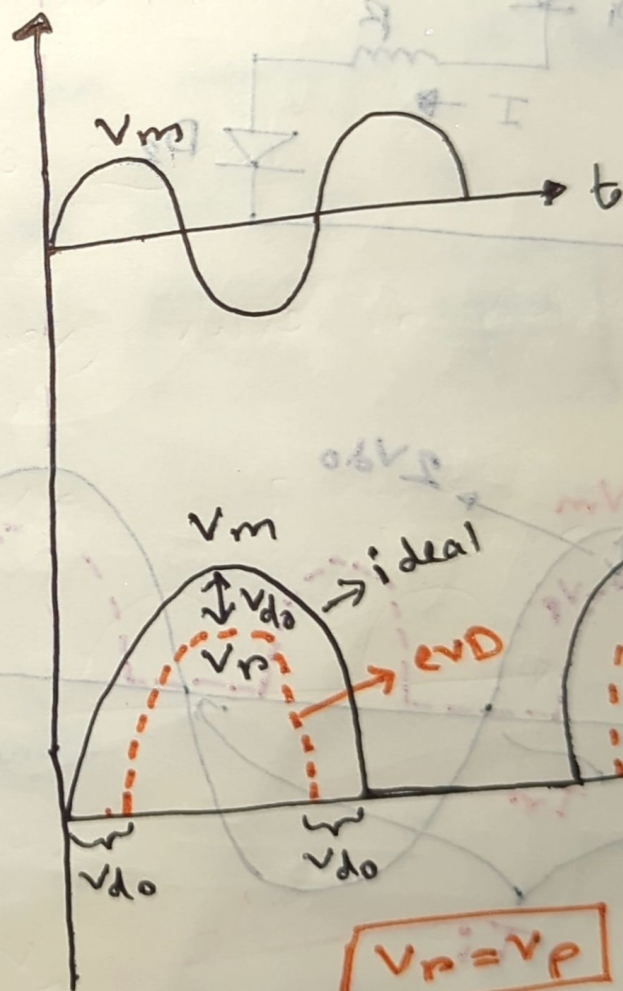
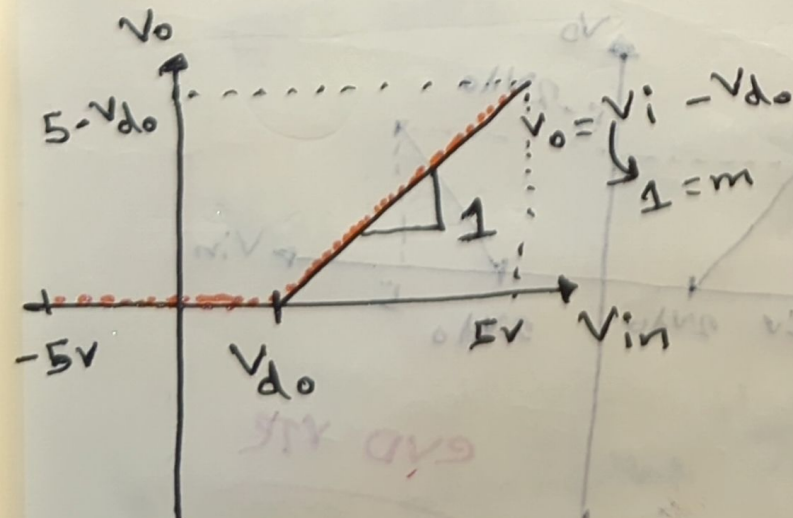
EVD



$$V_p = V_m - 0.7$$

$$V_p = V_m - V_{do}$$

VTE

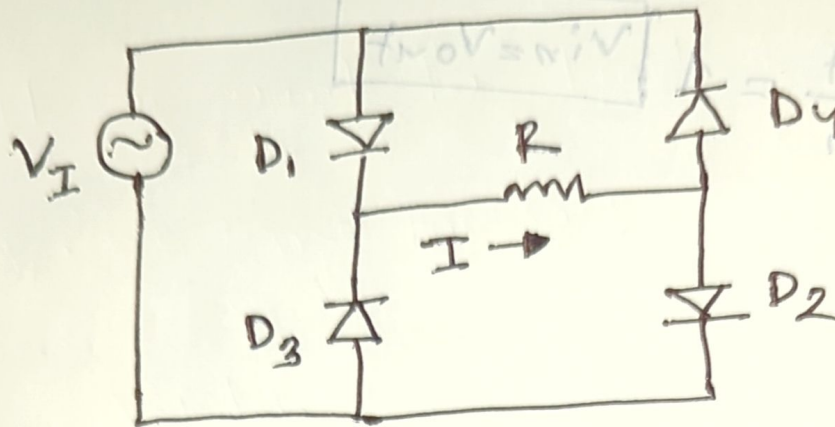


$$V_p = V_p$$

$$T_o = T_i$$

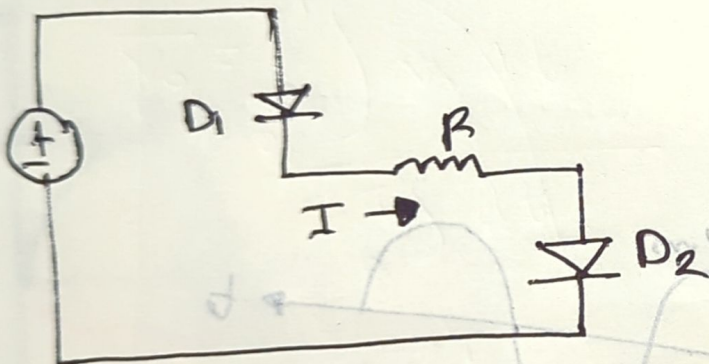
$$f_o = f_i$$

F.W

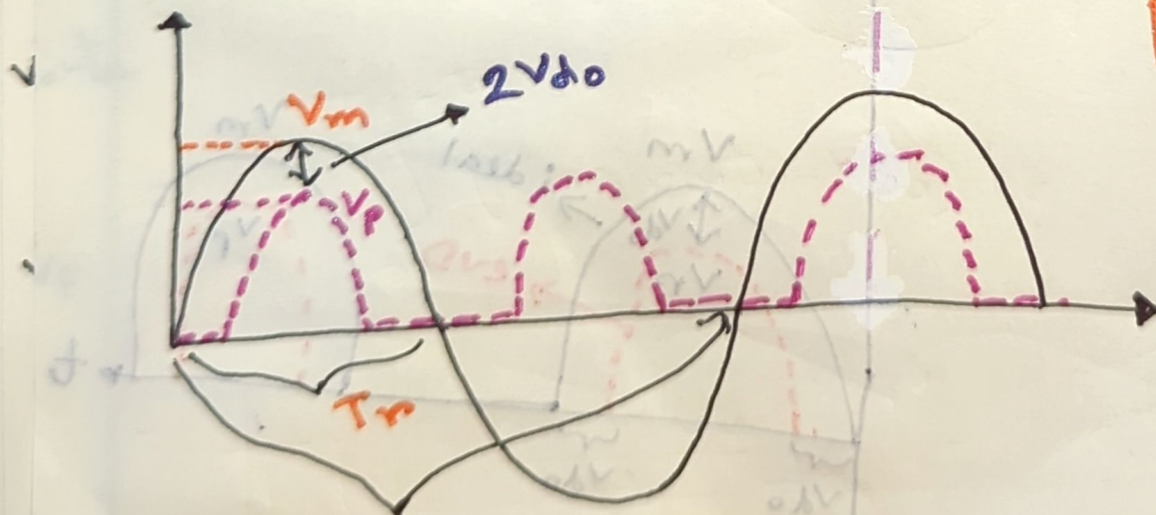
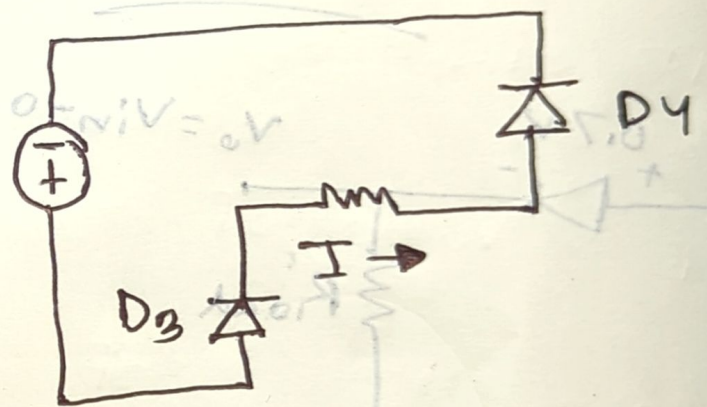


	+ve	-ve
D ₁	ON	OFF
D ₂	OFF	OFF
D ₃	OFF	ON
D ₄	OFF	ON

V_I (+ half cycle)



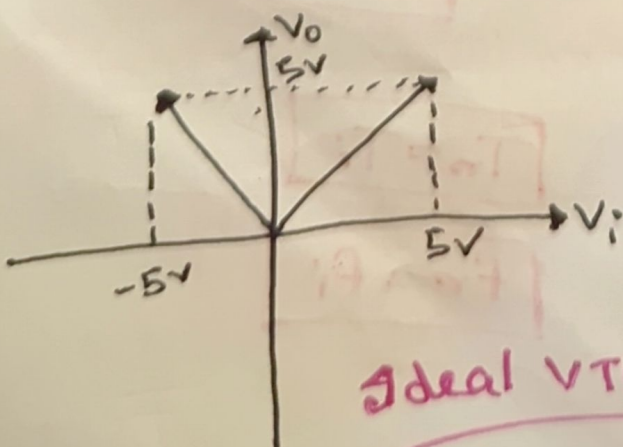
V_I (- half cycle)



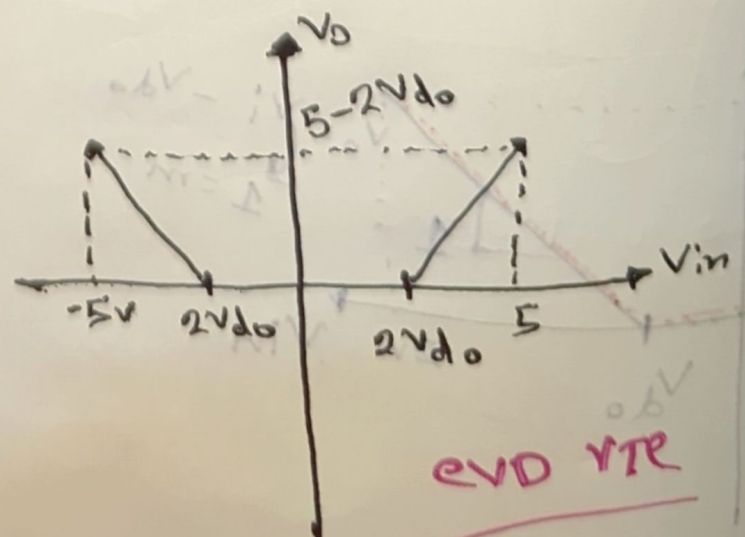
$$V_p = V_r$$

$$T_r = \frac{1}{2} T_i$$

$$f_r = 2f_i$$



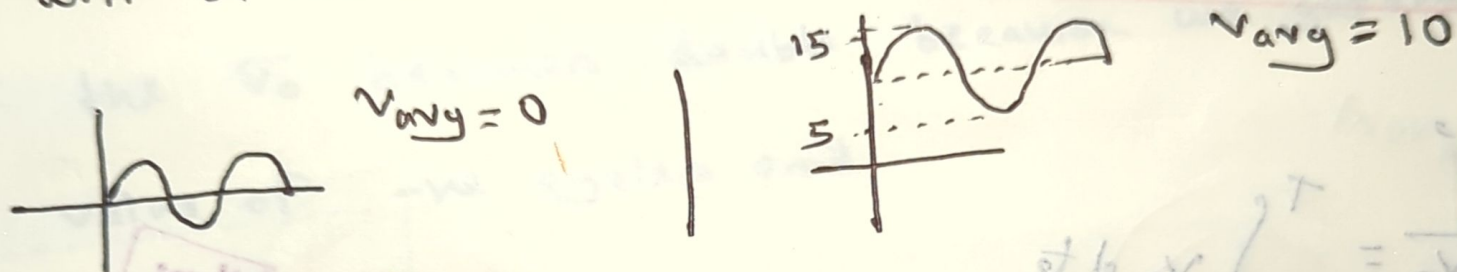
ideal VTE



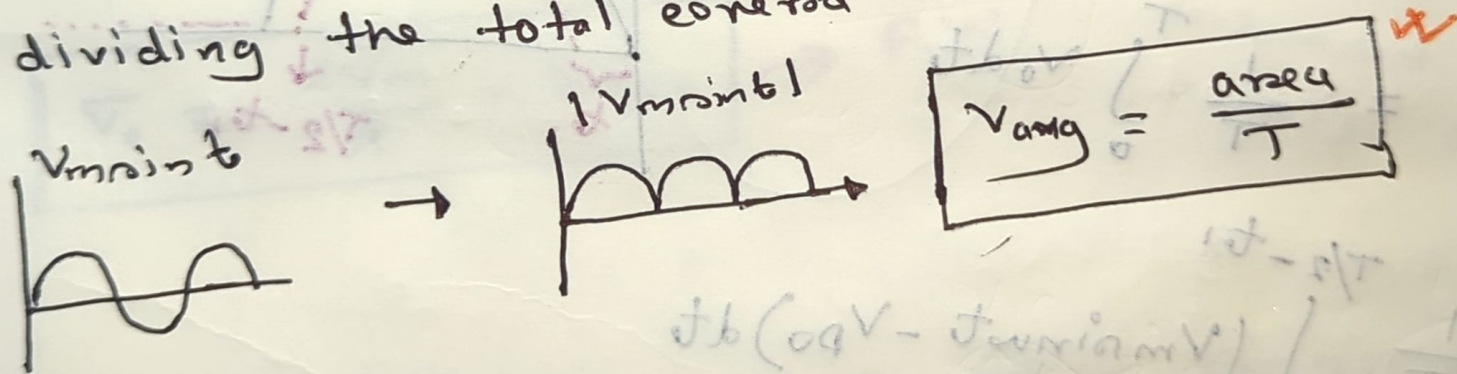
VD VTE

এখন আমরা $V_{dc} = V_{avg}$ এর বরাবর বিখ্যাত করে
আমাকে ultimately একটি constant voltage এ প্রকাশ
করতে হবে,

If we take avg of a pure AC signal, then it
will be zero means baseline basically.



But in pulsating DC we measure it by
dividing the total covered area by T.



আমরা এই area টা সূত্র Integration করে,
এর বরাবর,

$$V_{avg} = V_{dc} = \frac{V_m}{\pi} - \frac{V_{dc}}{2}$$

$$V_{avg} = V_{dc} = \frac{2V_m}{\pi} - 2V_{dc}$$

F.W

* From where the π comes dude. (???) $\rightarrow$ Let's see

Let's assume $y = f(x)$ is a function of x .
 So, if we want to find the average value of
 of y . We have to do:

$$\bar{y} = \frac{\int y dx}{\int dx}$$

for,

$$\bar{v}_o = \frac{\int_0^T v_o dt}{\int_0^T dt}$$

$$\bar{v}_o = \frac{1}{T} \int_0^T v_o dt$$

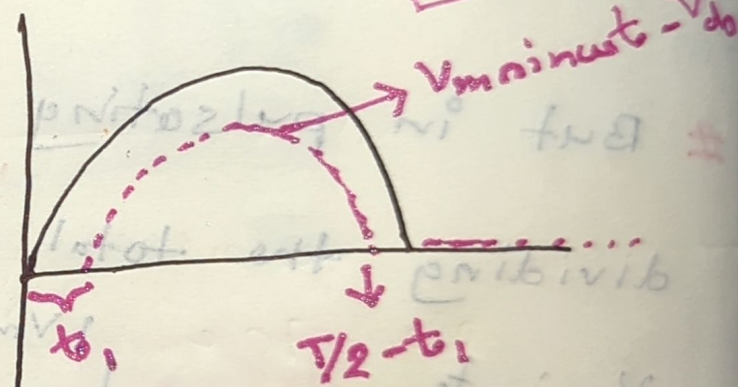
$$= \frac{1}{T} \int_{t_1}^{T/2 - t_1} (V_m \sin \omega t - V_{DO}) dt$$

$$= \frac{1}{T} \left\{ \left[-\frac{V_m \cos \omega t}{\omega} \right]_{t_1}^{T/2 - t_1} - V_{DO} [t]_{t_1}^{T/2 - t_1} \right\}$$

$$= \frac{1}{T} \left\{ -\frac{V_m}{\omega} [\cos \omega T/2 - \cos \omega 0] - V_{DO} [T/2 - 0] \right\}$$

[assuming $t_1 = 0$]

$$= \frac{1}{T} \left\{ -\frac{V_m}{\omega} \left(\cos \frac{2\pi}{T} \cdot \frac{T}{2} - 1 \right) - \frac{1}{T} V_{DO} \cdot \frac{T}{2} \right\}$$



$$\overline{V_o} = \frac{1}{\pi} \times \frac{V_m}{\frac{2\pi}{T}} \times 2 - \frac{V_{D0}}{2}$$

$$\overline{V_o} = \frac{V_m}{\pi} - \frac{V_{D0}}{2} = V_{D0} \rightarrow \text{H.W.}$$

for full wave the V_{D0} becomes double and the $\overline{V_o}$ becomes double because we get the value of -ve cycles end.

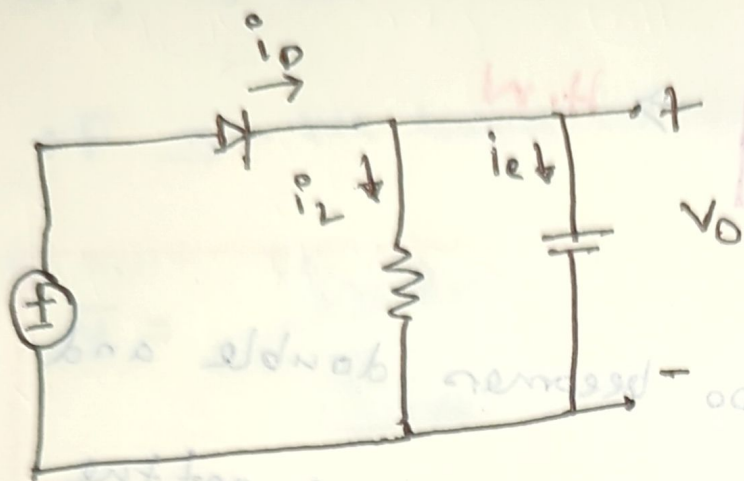
$$\overline{V_o} = 2 \left(\frac{V_m}{\pi} - \frac{2V_{D0}}{2} \right)$$

$$\overline{V_o} = \frac{2V_m}{\pi} - 2V_{D0} \rightarrow \text{F.W.}$$

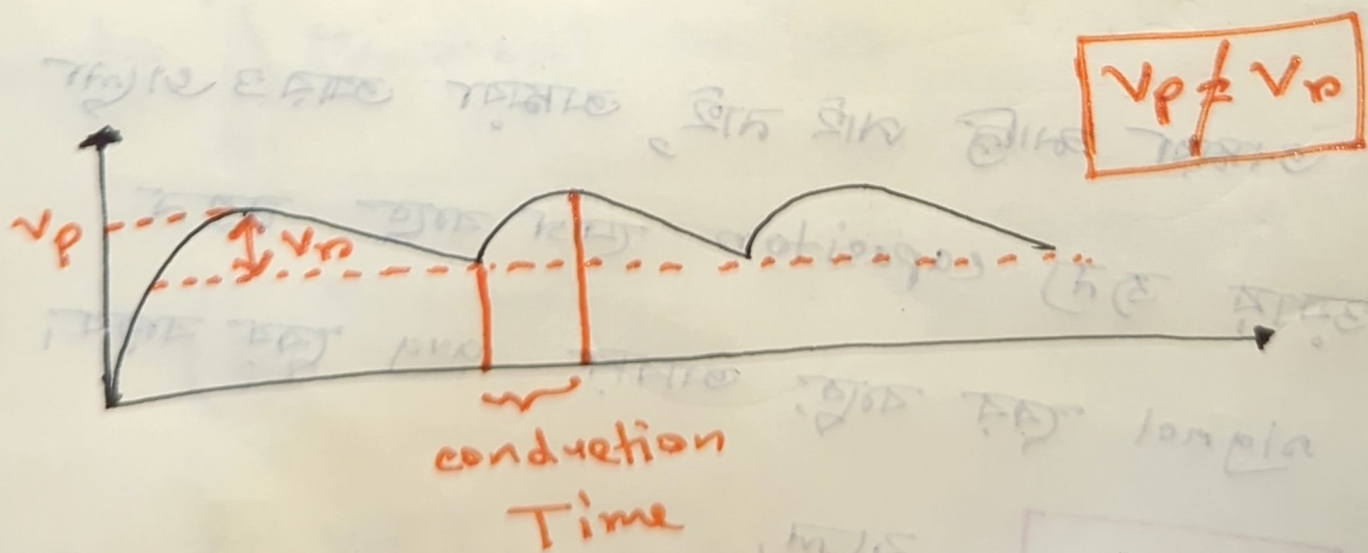
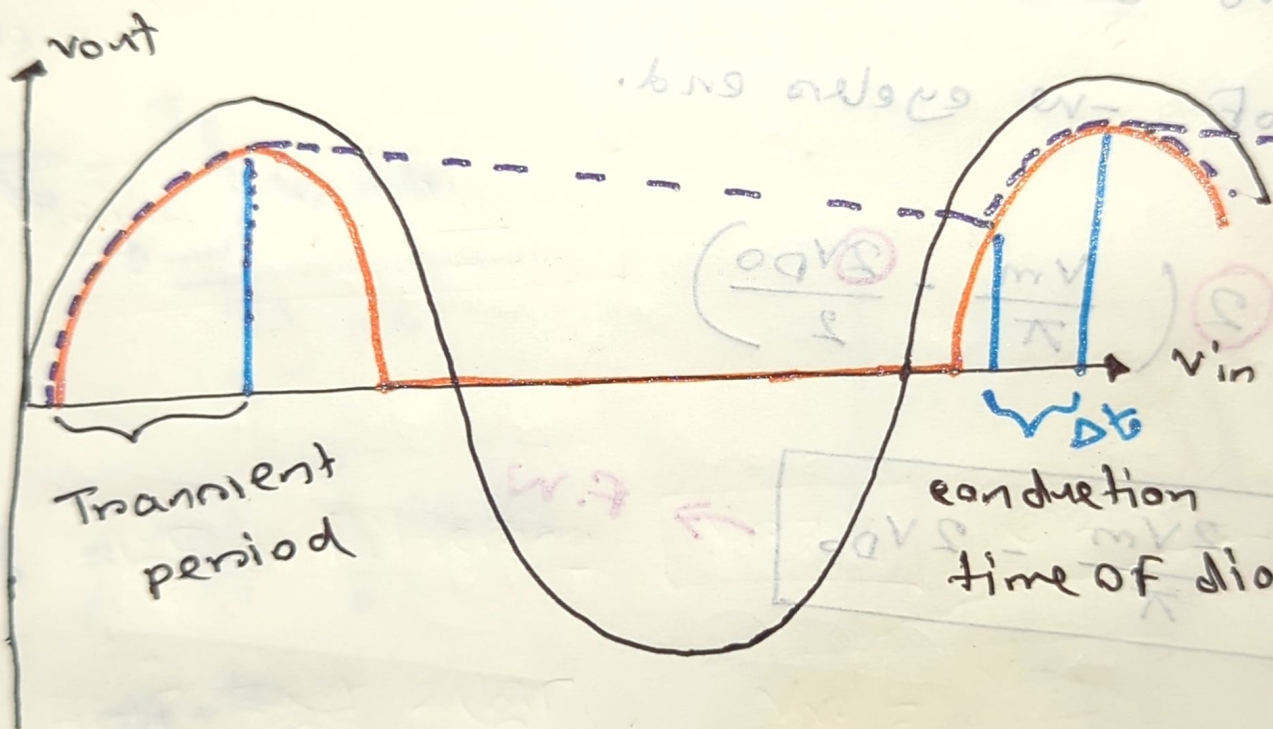
এখনও আমরা জানি না, আমরা আরও ভালো avg মানিয়ে কত capacitor লাগে হবে, নতুন output signal হবে, আমরা avg হবে, বাকি,

এটা **filtering** বলে,

Filtering



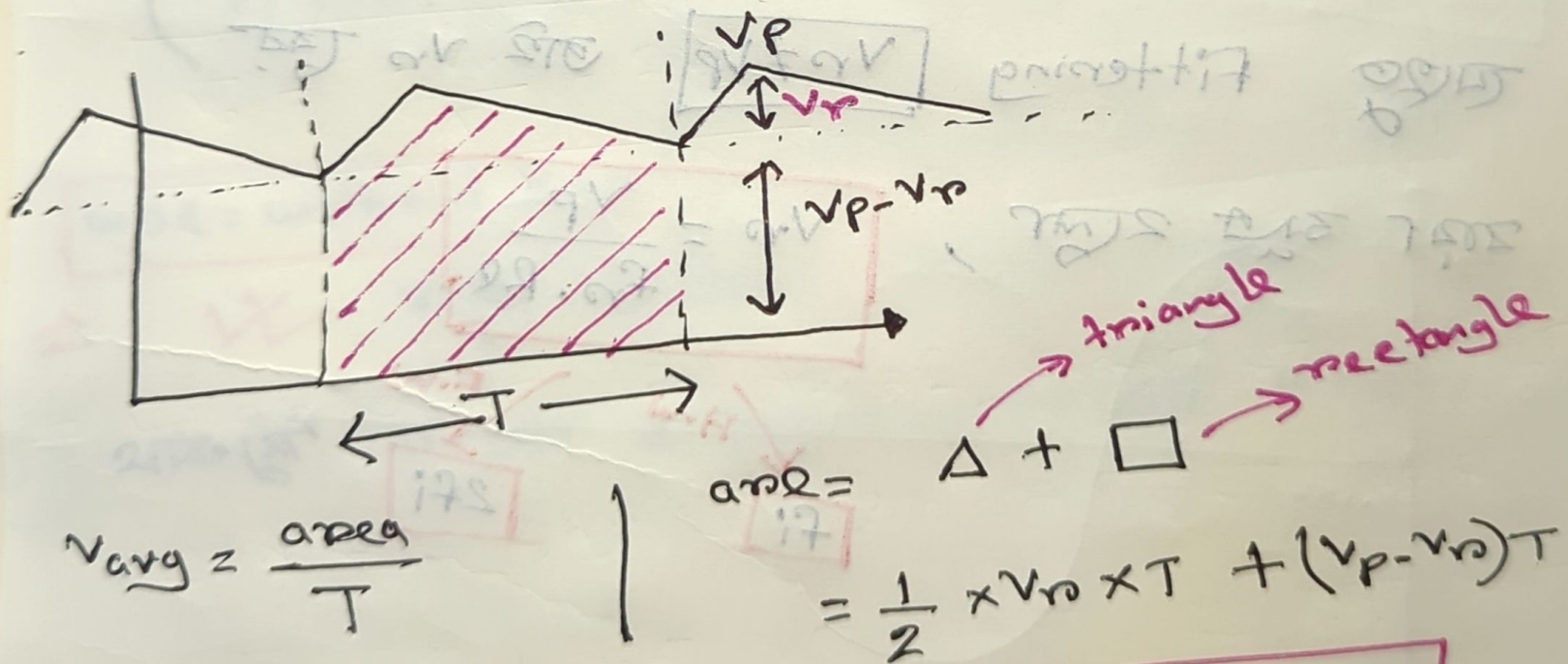
$$V_0 = \frac{V_m}{2} - \frac{V_m}{\pi} = \frac{V_m}{2}$$



ଆମର voltage ସହ peak ହାତେ ନାହିଁ, ତାହା capacitor ଓ ଅନ୍ୟ କଣ୍ଡକ୍ତର voltage ହେବ, ତାହା voltage ଆମର ଆମର ନିକଟତର ହେବ

আর প্রযুক্তি Transient period capacitor চার্জ
 হয় তারপর যাকি অব অসম্পূর্ণ conduction
 time এ চার্জ হয়, Diode যুক্ত এই অসম্পূর্ণ চার্জ
 বহন করে। আর ন conduction time (কিন্তু
 হওয়ায় পর Diode এর -এ প্রচুর voltage আর
 capacitor এর কারণে এর + চার্জ আর
 voltage থাকে।

finding the avg: (taking an a triangle signal)
 to calculate precisely



$$V_{avg} = \frac{V_r}{2} + V_p - V_r$$

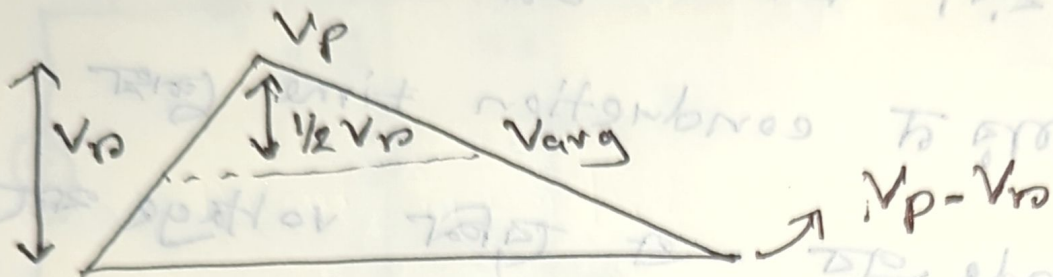
$$\text{area} = \frac{V_r T}{2} + (V_p - V_r)T$$

$$V_{avg} = V_p - \frac{V_r}{2}$$

H.W.

ଆହୁରି ଏକଟି ଭାବ୍ଯ ଡିମାଣ୍ଡ ଉପରେ ମାରି ହୁଏ

ସୂତ୍ର ଅଟେ - ଅନ୍ୟତମ ବା mid value



$$\text{so } V_{avg} = \frac{V_p + (V_p - V_n)}{2} = \frac{2V_p - V_n}{2} = V_p - \frac{V_n}{2}$$

$$V_{avg} = V_p - \frac{V_n}{2}$$

ଯଦି Fitting $V_n \neq V_p$ ତାହା V_n ହୁଏ

ଆମେ ଏହା ଧ୍ୟାନ ଦେବା,

$$V_n = \frac{V_p}{f_r \cdot R_e}$$

H.W. f_i

F.W. $2f_i$

Formulla

M.H.

$$f_n = f_i$$

$$V_p = V_m - V_{do}$$

$$V_{de} = \begin{cases} \frac{V_m}{\pi} - \frac{1}{2} V_{do} [\omega \theta_e] \\ V_p - \frac{1}{2} V_r [\omega \theta_e] \end{cases}$$

$$v_p = \begin{cases} v_p(\omega_{ce}) \\ v_p / f_{nRE}(\omega_{ce}) \end{cases}$$

F.V

$$f_{ro} = 2f; \quad \checkmark$$

$$V_p = V_m - 2V_{do}$$

$$V_{de} = \begin{cases} \frac{2V_m}{\pi} - 2V_{do} & [\omega o e] \\ V_p - \frac{1}{2} V_r & [\omega e] \end{cases}$$

$$V_p = \begin{bmatrix} V_p & [w_{oe}] \\ V_p / f_p R_L & [w_e] \end{bmatrix}$$

woe = without eap

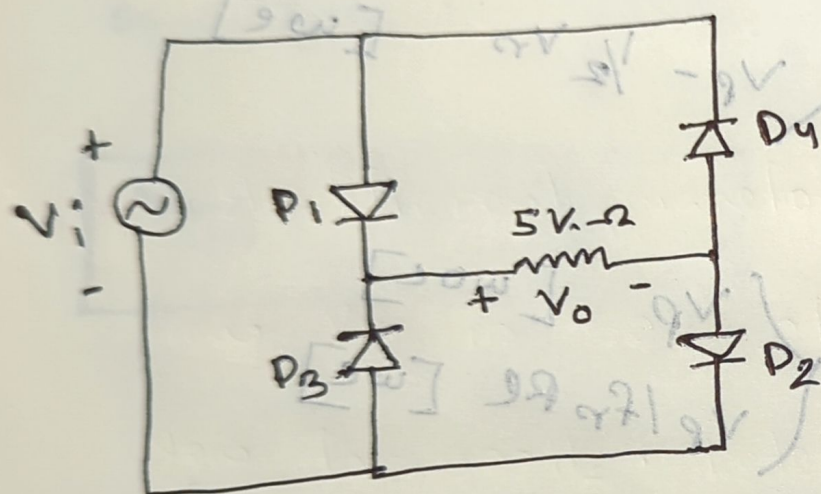
we = with ear

14 ସଂଯୁକ୍ତ ତ୍ରିଭିନ୍ନ ଶୂନ୍ୟ capacitor
ଆବୃତ୍ତି ବା ନା ପାସାଲେ 3 ପ୍ରକାରର ଥାନ୍ତି.

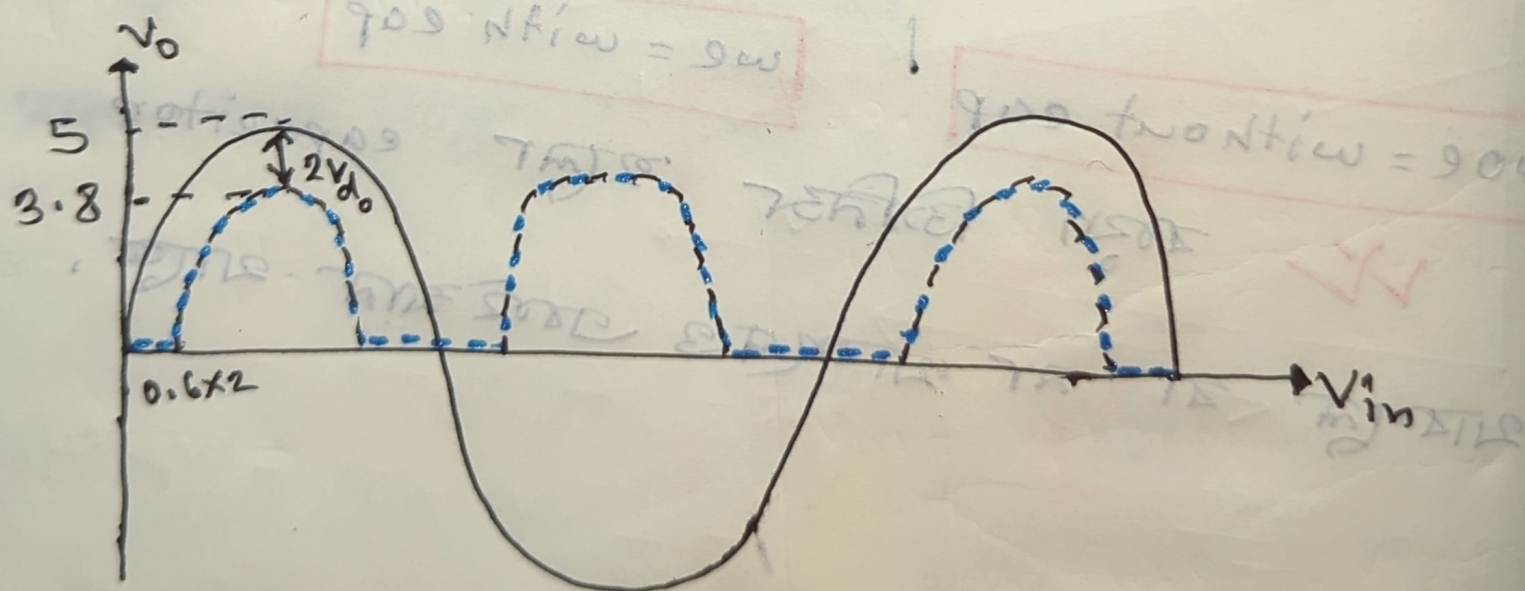
Diode Rectifier Problem

1) $V_i = 5 \sin(200\pi t)$, $R = 5k\Omega$, $V_{D0} = 0.6$ Full wave

Ans: Draw circuit, Label input and output voltage
Briefly explain the application of the circuit.



	the	-ve
D ₁	ON	OFF
D ₂	ON	OFF
D ₃	OFF	ON
D ₄	OFF	ON



BQ: calculate V_{de} and output frequency f_o .

Given, $V_i = 5 \sin(200\pi t)$ comparing to

$$V_i = V_m \sin(2\pi f_i t)$$

$$f_i = 200/2 = 100 \text{ Hz}$$

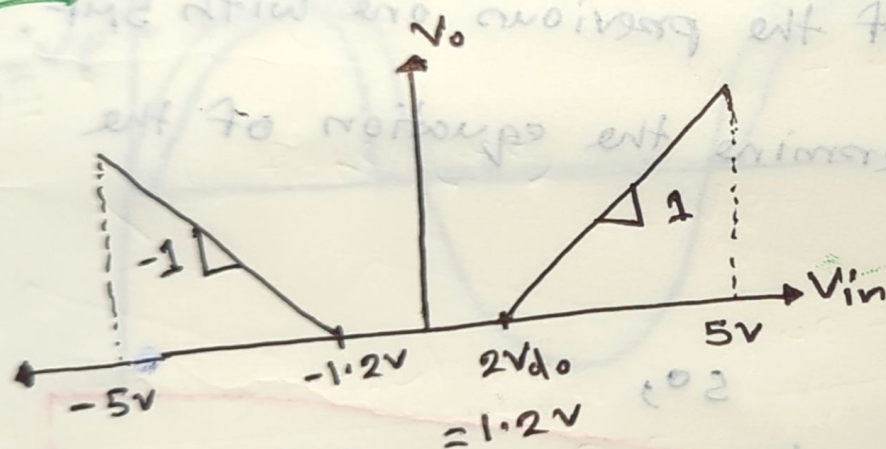
$$f_o = 2 \times 100 = 200 \text{ Hz} \quad [F.W \rightarrow f_o = 2f_i]$$

$$V_{de} = \frac{2V_m}{\pi} - 2V_{d0}$$

$$V_{de} = \frac{2 \times 5}{\pi} - 2 \times 0.6$$

$$V_{de} = 1.583 \text{ V}$$

eg: Draw VTE and Label it.



DQ: connect a capacitor with the circuit parallelled and we have two options to choose $5\mu\text{F}$ / $1\mu\text{F}$. Which one will you choose explain briefly.

when, $5\mu\text{F}$,

$$V_{de} = V_p - \frac{1}{2} V_r$$

$$= V_m - 2V_{d0} - \frac{V_p}{2f_o R C}$$

$$= V_m - 2V_{d0} - \frac{V_m - 2V_{d0}}{2f_o R C}$$

$$V_p = V_m - 2V_{d0}$$

$$V_r = \frac{V_p}{f_o R C}$$

$$f_o = f_i = 200 \text{ Hz} \quad | \quad V_m = 5 \text{ V}$$

$$R = 5 \text{ k}\Omega$$

$$C = 5 \mu\text{F}$$

$$V_{de} = 5 - 2 \times 0.6 - \frac{5 - 2 \times 0.6}{2 \times 200 \times 5 \times 10^3 \times 5 \times 10^{-6}}$$

$$V_{de} = 3.42V$$

when, $1\mu F$

$$V_{de} = 1.9V$$

so, to get the highest V_{de}

we will use $5\mu F$

Ex: A different waveform fed into f.w. The new V_r is 50% of the previous one with $5\mu F$. New $f_0 = 300Hz$. Determine the equation of the input waveform.

Given,

$$f_0 = 300Hz$$

$$V_r = \frac{V_p}{f_0 R E} / 2$$

$$= \frac{5 - 2 \times 0.6}{2 \times 200 \times 5 \times 10^3 \times 5 \times 10^{-6}}$$

$$V_r = 0.38V$$

$$f_i = f_0 / 2 = \frac{300}{2} = 150Hz$$

$$V_m = V_p + 2V_{d0}$$

$$= V_r f_0 R E + 2V_{d0}$$

$$= 0.38 \times 300 \times 5 \times 10^3 \times 5 \times 10^{-6} + 2 \times 0.6$$

$$V_m = 4.05V$$

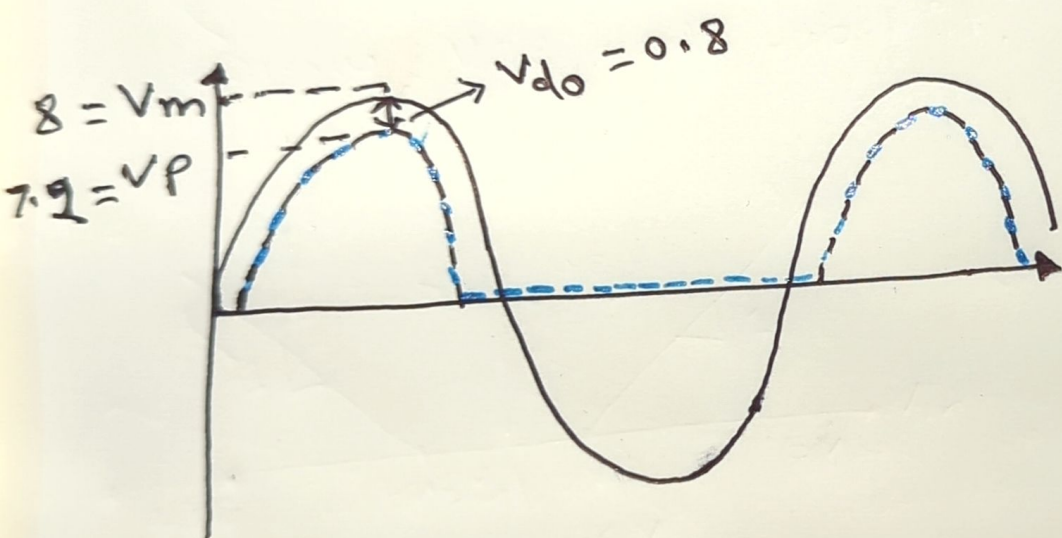
$$V_i = 4.05 \sin(300\pi t) V$$

2 $V_i = 8 \sin(2000\pi t) \text{ V}$, H.W, $R = 50 \text{ k}\Omega$

$V_{do} = 0.8 \text{ V}$

AG: same as 1 AG $\rightarrow$ asked to do it in F.W.

BB: Draw waveforms for input and output voltages.
Label peak values.



eg: Find V_{de} .

$$V_{de} = \frac{V_m}{\pi} - \frac{V_{do}}{2}$$

$$V_{de} = \frac{8}{\pi} - \frac{0.8}{2}$$

$V_{de} = 2.146$